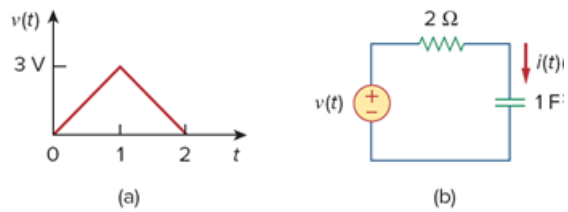


Problem

Determine the current $i(t)$ in the circuit of Fig. 18.42(b), given the voltage source shown in Fig. 18.42(a).

Figure 18.42



Step-by-step solution

Step 1 of 9

(Refer to Figure 18.42 (a) in the textbook) the voltage $v(t)$ is represented as,

$$v(t) = \begin{cases} 3t, & 0 < t < 1 \\ 6 - 3t, & 1 < t < 2 \end{cases}$$

Step 2 of 9

Take the first derivative of the signal $v(t)$.

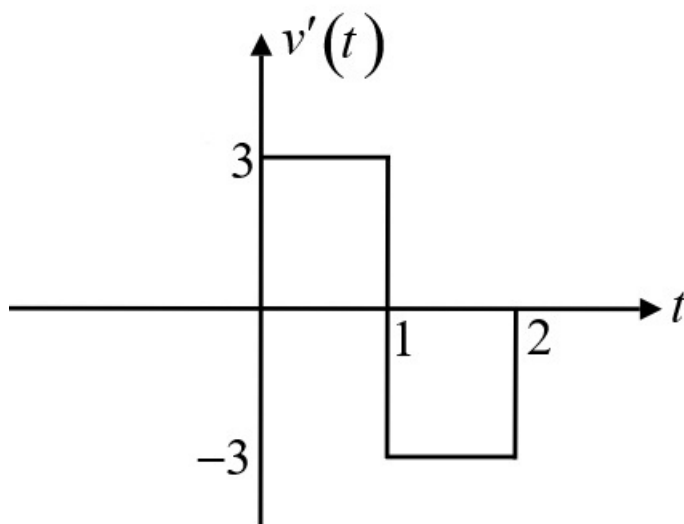


Figure 1

Step 3 of 9

Take the second derivative of signal $v(t)$.

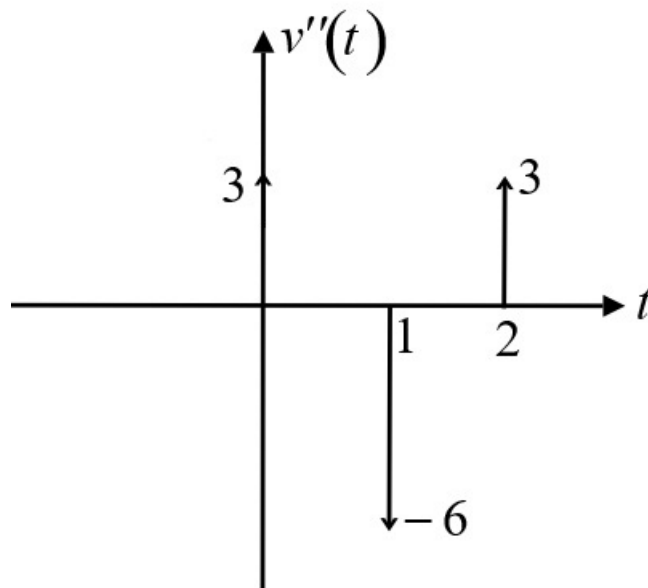


Figure 2

Step 4 of 9

The second derivative of the signal $v''(t)$ is represented as,

$$v''(t) = 3\delta(t) - 6\delta(t-1) + 3\delta(t-2)$$

Take the Fourier transform.

$$v''(\omega) = 3 - 6e^{-j\omega} + 3e^{-2j\omega}$$

By the derivative property of the Fourier transform,

$$\begin{aligned} V(\omega) &= \frac{v''(\omega)}{(j\omega)^2} \\ &= \frac{3 - 6e^{-j\omega} + 3e^{-2j\omega}}{(j\omega)^2} \end{aligned}$$

Step 5 of 9

Represent the circuit in frequency domain (Refer Figure 18.42 (b) in the textbook).

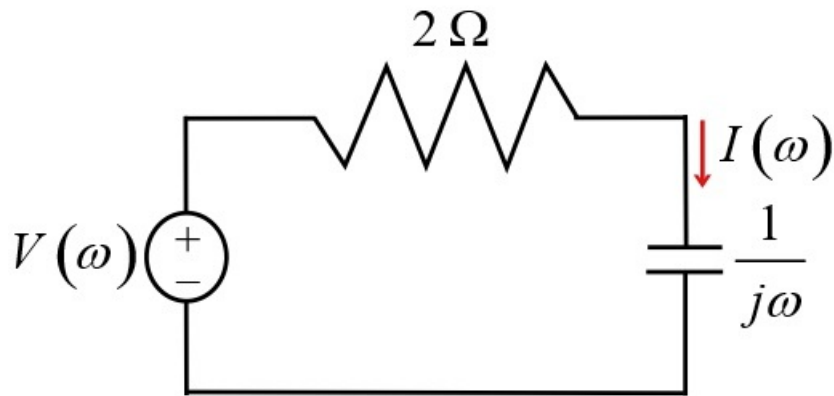


Figure 3

Step 6 of 9

Apply the Kirchhoff's voltage law.

$$V(\omega) = \left(2 + \frac{1}{j\omega} \right) I(\omega)$$

Substitute $\frac{3 - 6e^{-j\omega} + 3e^{-2j\omega}}{(j\omega)^2}$ for $V(\omega)$.

$$\frac{3 - 6e^{-j\omega} + 3e^{-2j\omega}}{(j\omega)^2} = \left(2 + \frac{1}{j\omega} \right) I(\omega)$$

$$I(\omega) = \frac{1}{(j\omega)(0.5 + j\omega)} \left[\frac{3}{2} - 3e^{-j\omega} + \frac{3e^{-2j\omega}}{2} \right]$$

Step 7 of 9

Calculate the partial fraction.

$$\frac{1}{(j\omega)(0.5 + j\omega)} = \frac{A}{j\omega} + \frac{B}{0.5 + j\omega} \dots\dots (1)$$

$$1 = A(0.5 + j\omega) + B(j\omega)$$

Substitute 2 for A and -2 for B in equation (1).

$$\frac{1}{(j\omega)(0.5 + j\omega)} = \frac{2}{j\omega} - \frac{2}{0.5 + j\omega}$$

Step 8 of 9

Then, the current $I(\omega)$ is,

$$\begin{aligned} I(\omega) &= \left[\frac{2}{j\omega} - \frac{2}{0.5 + j\omega} \right] \left[\frac{3}{2} - 3e^{-j\omega} + \frac{3e^{-2j\omega}}{2} \right] \\ &= \frac{3}{2} \left[\frac{2}{j\omega} \right] - 3 \left[\frac{2e^{-j\omega}}{j\omega} \right] + \frac{3}{2} \left[\frac{2e^{-2j\omega}}{j\omega} \right] - \frac{3}{0.5 + j\omega} + \frac{6e^{-j\omega}}{0.5 + j\omega} - \frac{3e^{-2j\omega}}{0.5 + j\omega} \end{aligned}$$

Step 9 of 9

Calculate the inverse Fourier transform.

$$i(t) = \frac{3}{2} \operatorname{sgn}(t) - 3 \operatorname{sgn}(t-1) + \frac{3}{2} \operatorname{sgn}(t-2) - 3e^{-0.5t} u(t) + 6e^{-0.5(t-1)} u(t-1) - 3e^{-0.5(t-2)} u(t-2)$$

Hence, the current $i(t)$ in the circuit is

$$\boxed{\frac{3}{2} \operatorname{sgn}(t) - 3 \operatorname{sgn}(t-1) + \frac{3}{2} \operatorname{sgn}(t-2) - 3e^{-0.5t} u(t) + 6e^{-0.5(t-1)} u(t-1) - 3e^{-0.5(t-2)} u(t-2) \text{ A}}$$